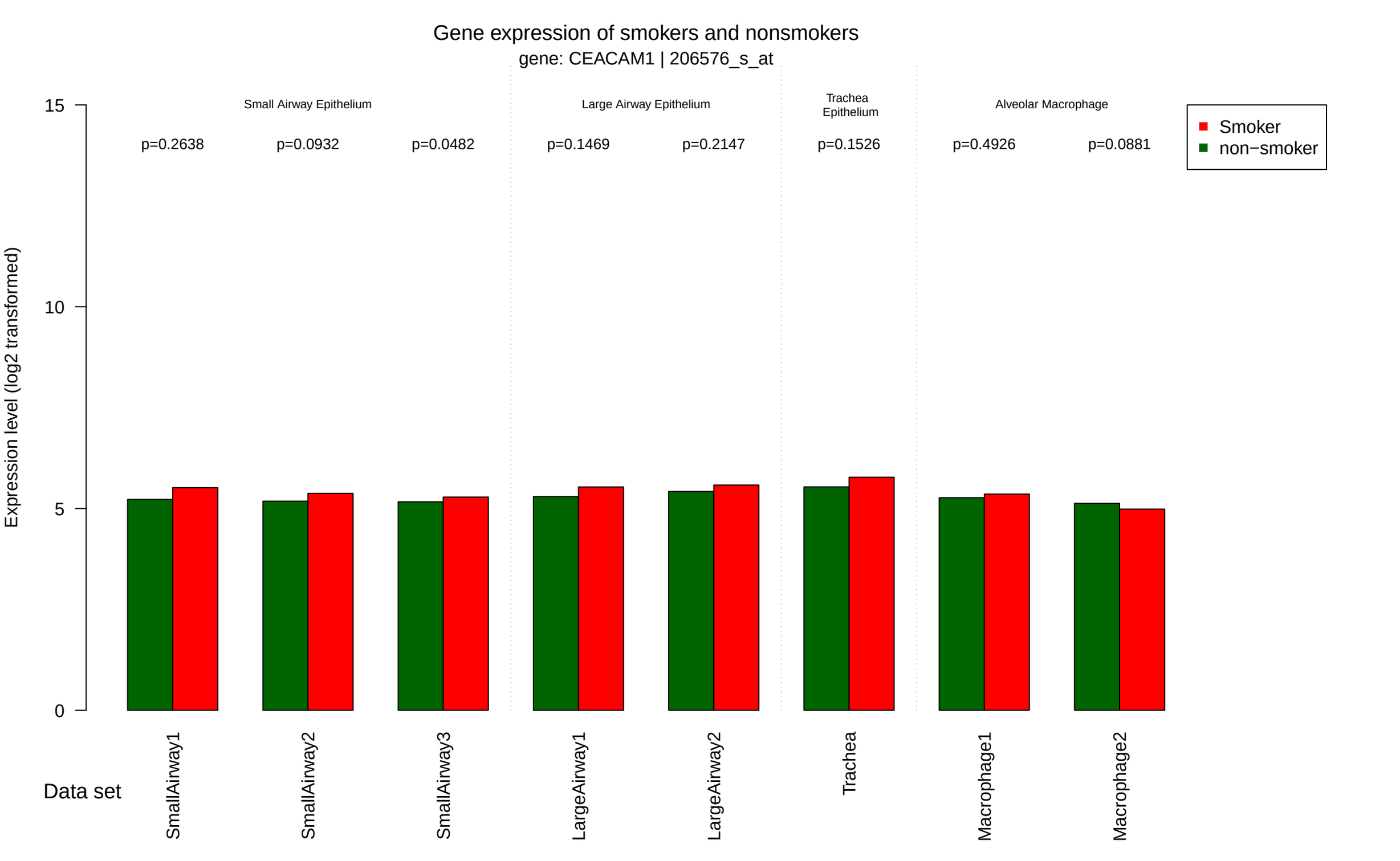
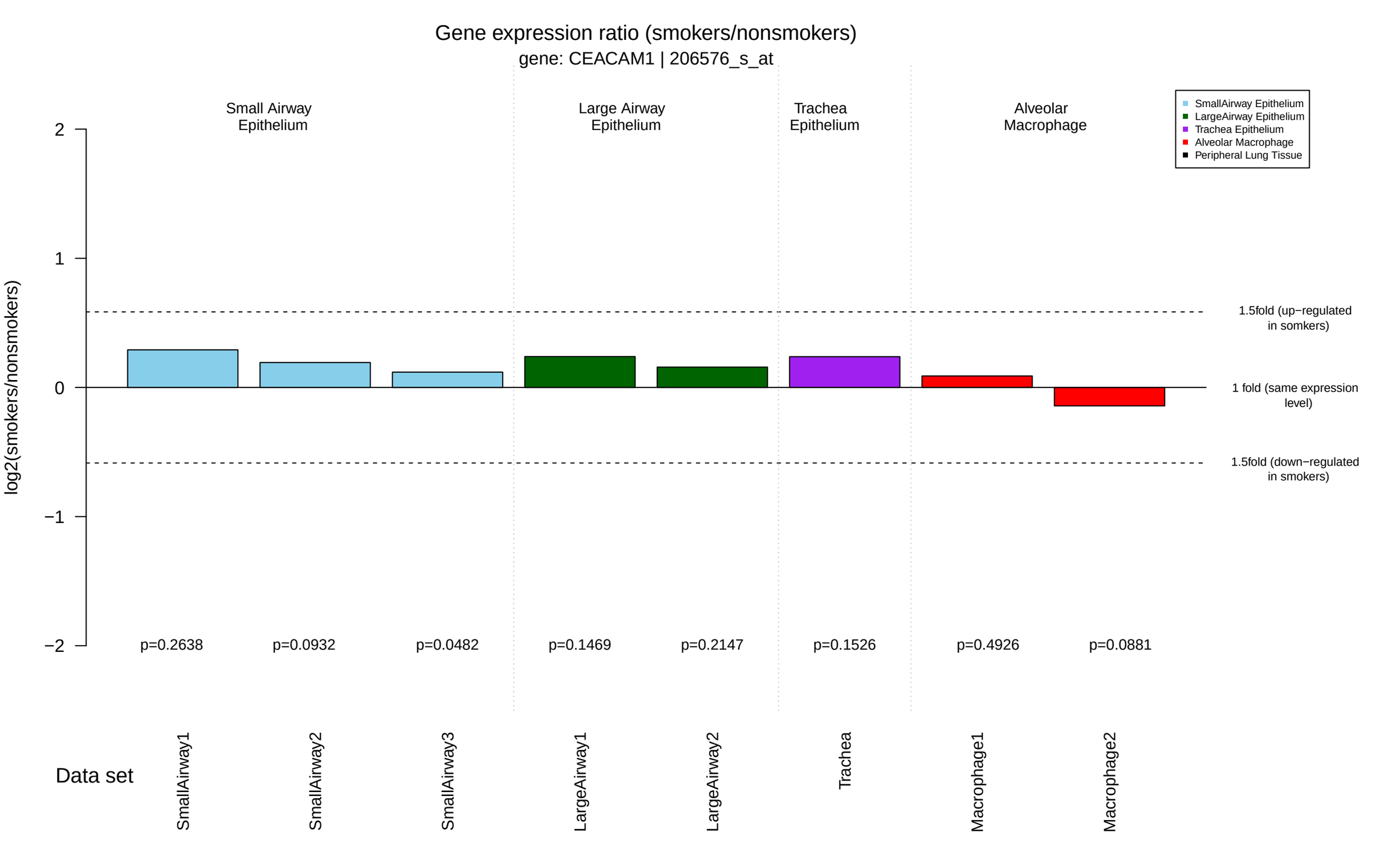
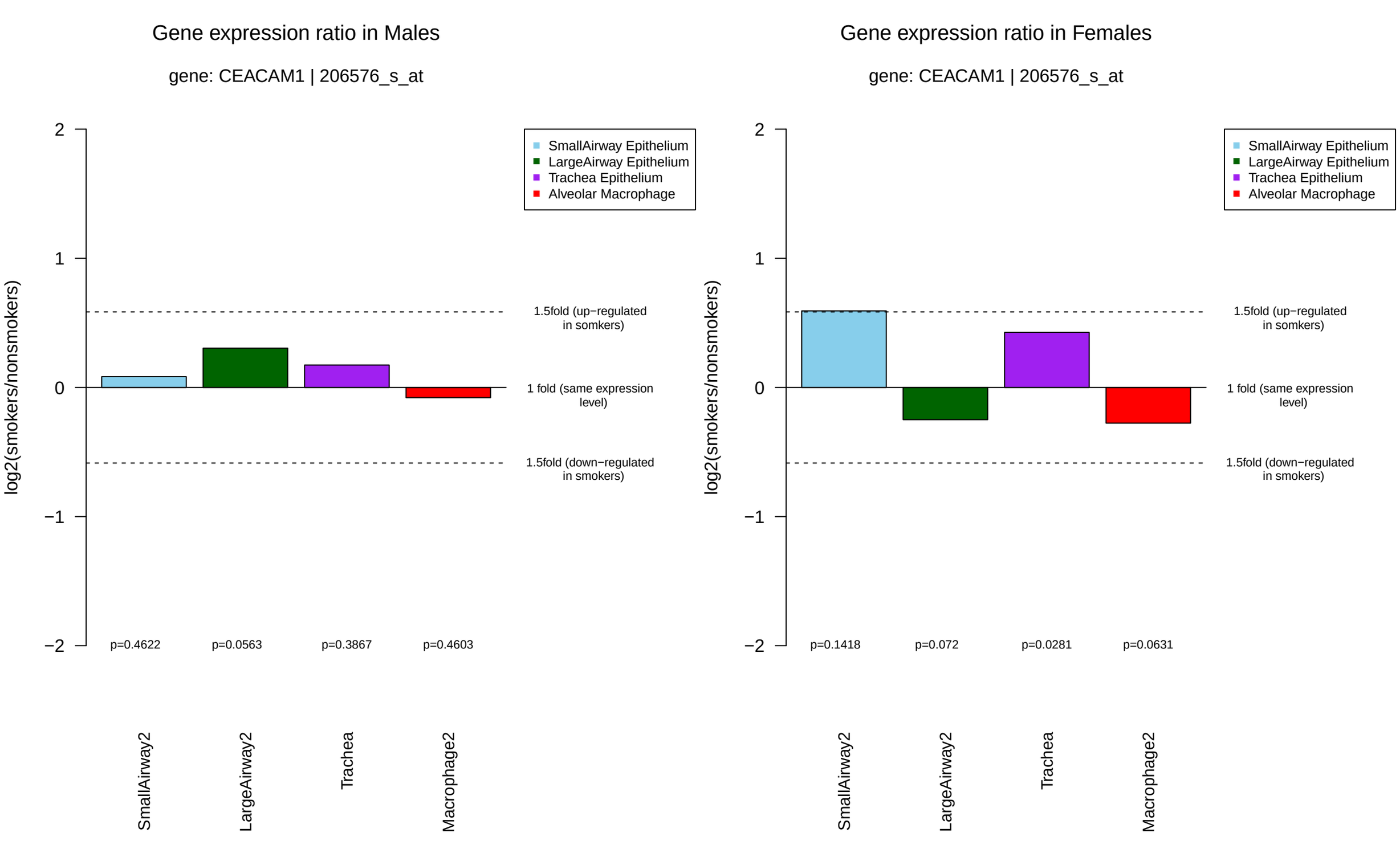
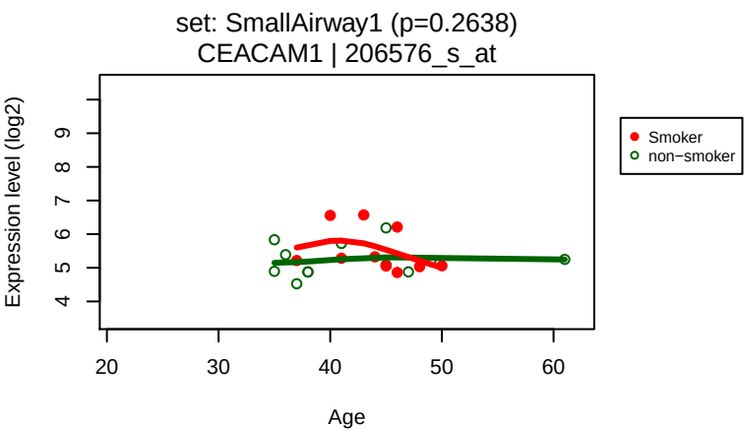
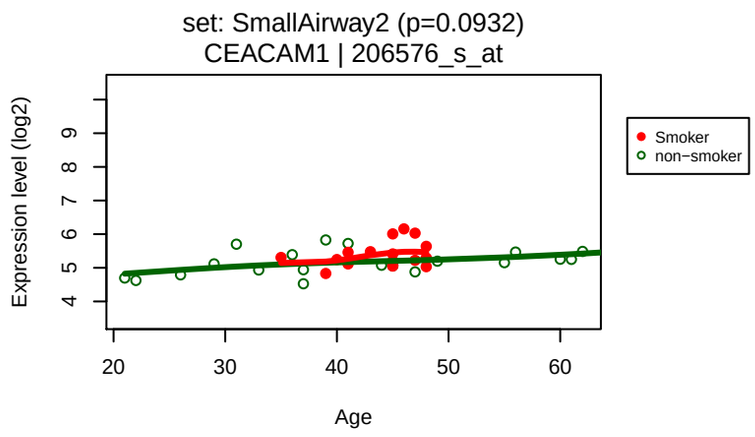
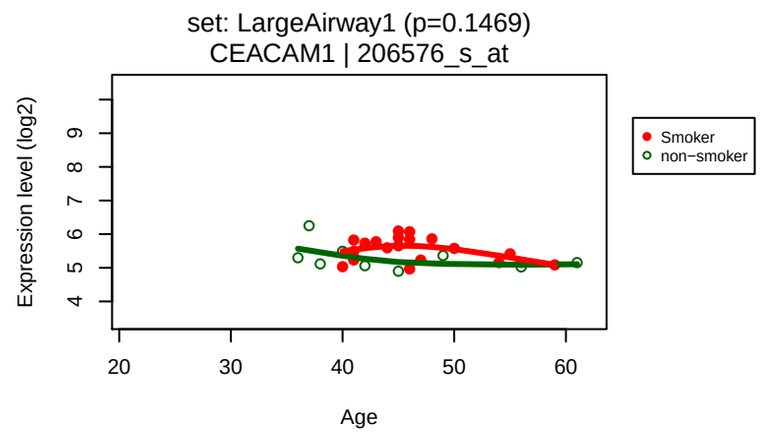

Gene expression of smokers and nonsmokers
gene: CEACAM1 | 206576_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2638
p=0.0932
p=0.0482
p=0.1469
p=0.2147
p=0.1526
p=0.4926
p=0.0881
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

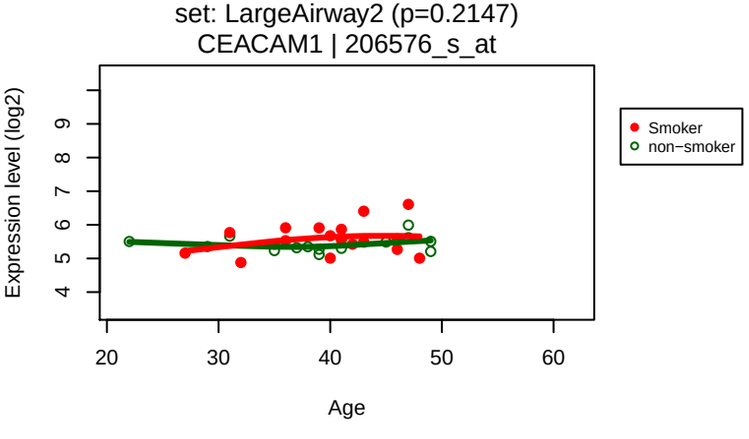

Gene expression ratio (smokers/nonsmokers)
gene: CEACAM1 | 206576_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2638
p=0.0932
p=0.0482
p=0.1469
p=0.2147
p=0.1526
p=0.4926
p=0.0881
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

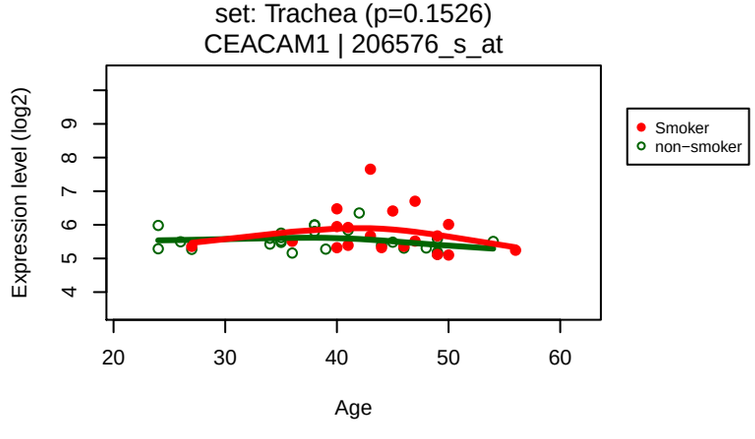

Gene expression ratio in Males
gene: CEACAM1 | 206576_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4622
p=0.0563
p=0.3867
p=0.4603
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CEACAM1 | 206576_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1418
p=0.072
p=0.0281
p=0.0631
SmallAirway2
LargeAirway2
Trachea
Macrophage2

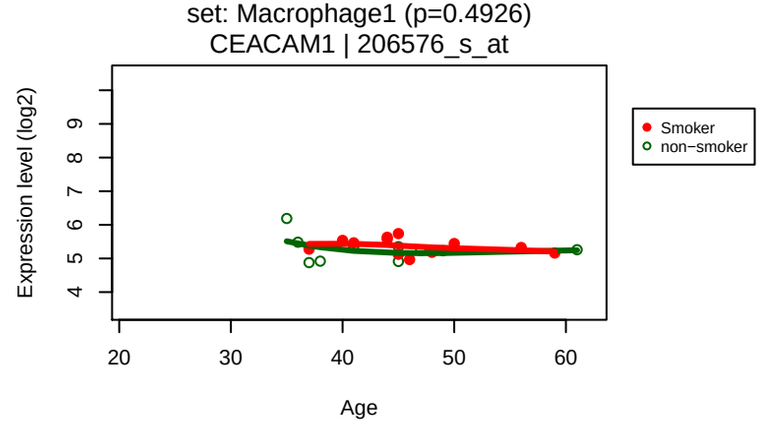

set: SmallAirway1 (p=0.2638)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker

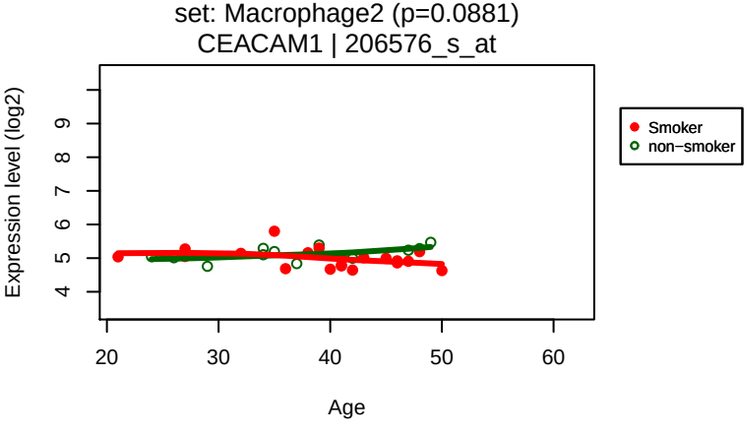

set: SmallAirway2 (p=0.0932)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker

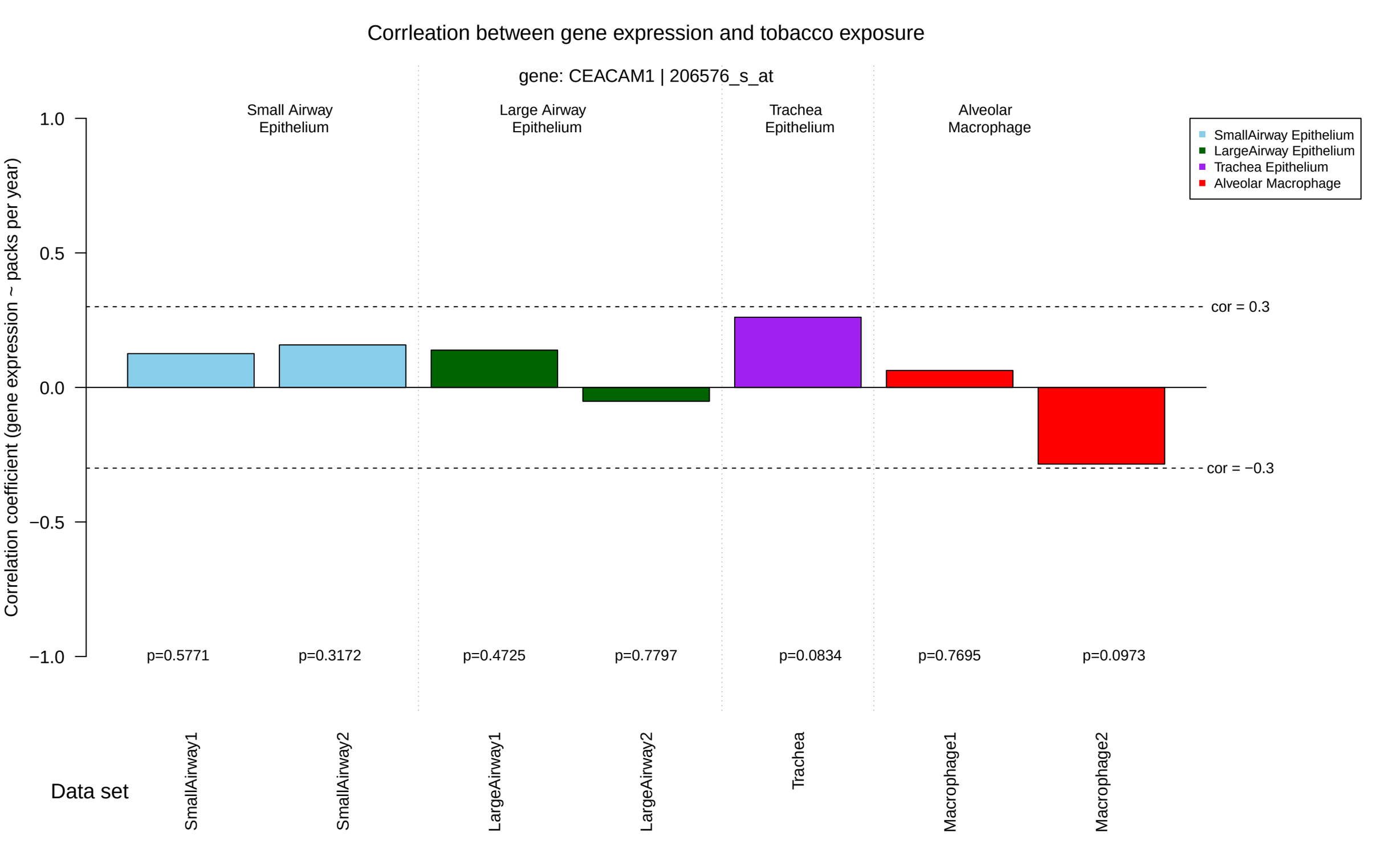

set: LargeAirway1 (p=0.1469)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2147)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1526)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4926)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0881)
CEACAM1 | 206576_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CEACAM1 | 206576_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5771
p=0.3172
p=0.4725
p=0.7797
p=0.0834
p=0.7695
p=0.0973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2